GATE-AII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 08



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Recape of previous lecture



Topic Maxima and minima of functions

Topic Problems based on maxima and minima

Topics to be covered



Topic Rolles theorem

Topic Lagrange mean value theorem

Topic Problems based on mean value theorem

Topic Basic integration

MEAN value Theorems:

Rolle's Theorem:

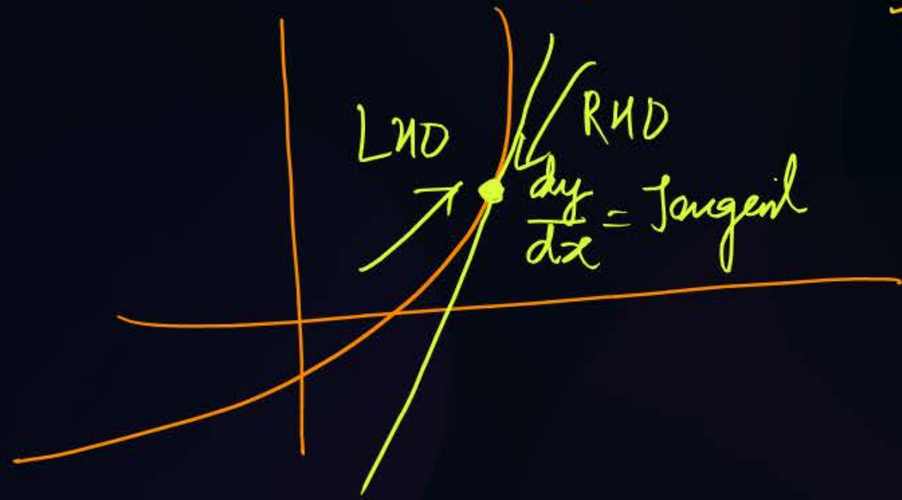
Differentiability $\Rightarrow$ LHD = RHD

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{+h} = \text{RHD}$$

$$\lim_{h\to 0} \frac{f(a-h)-f(a)}{-h} = \text{LHD}$$

Function must be Differentiable

Tangent $\frac{dy}{dx}$



1) $y = f(x)$

(A) $f(x)$ is Differentiable in (a,b)

2)

(B) $f(x)$ is continuous in $[a,b]$

LHL = RHL

$$\lim_{h\to 0} f(a+h) = \lim_{h\to 0} f(a-h)$$

(c) $\boxed{f(a) = f(b)}$

If three conditions are satisfied

Then at least one Point $\rightarrow \boxed{f'(c) = 0}$

$c \in (a, b)$

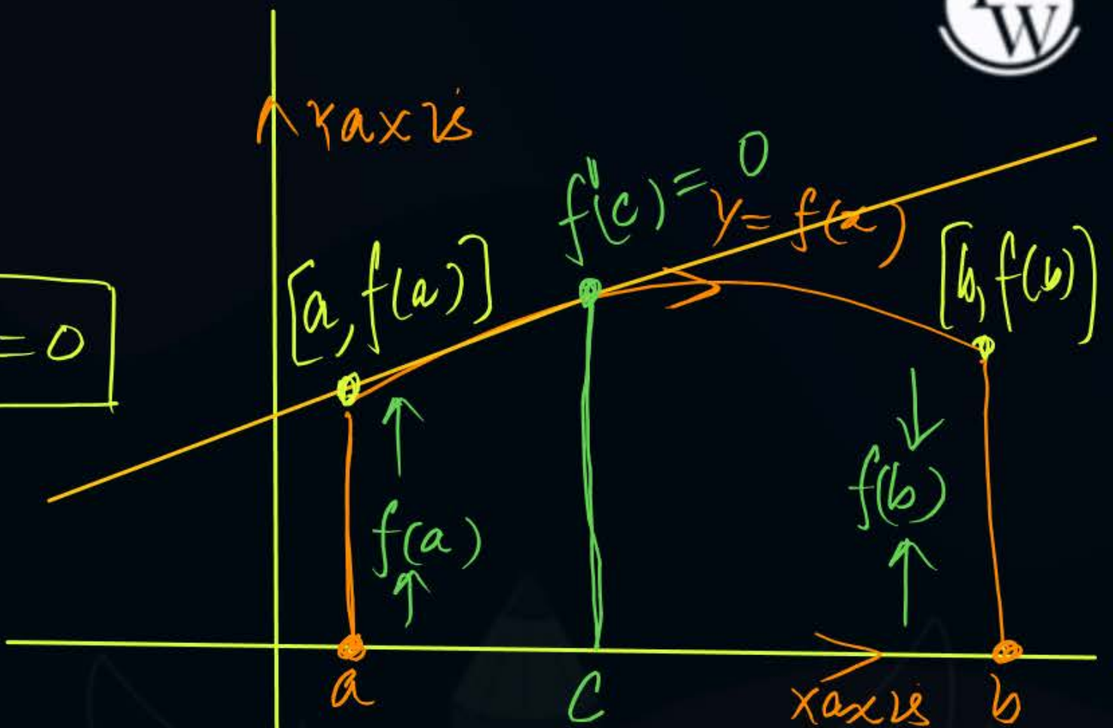
Proof: $y = f(x)$ given Points $(a, f(a))$ $(b, f(b))$

x_1, y_1 x_2, y_2

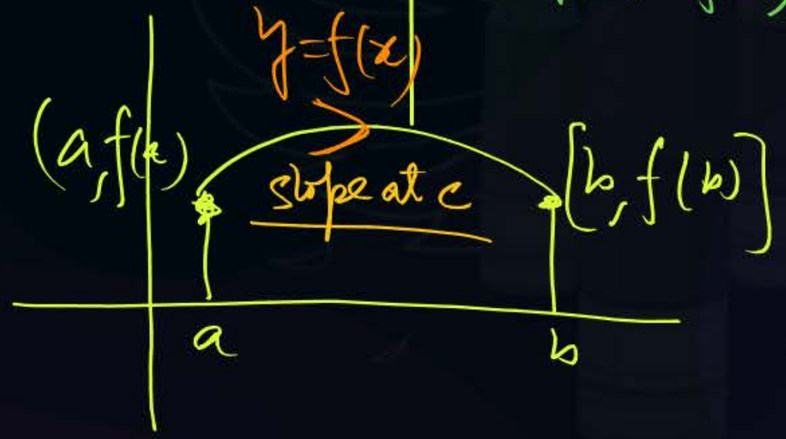
slope

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope} = \frac{f(b) - f(a)}{(b - a)}$$



$f(a) = f(b)$



$$\text{slope} = \frac{f(b)-f(a)}{(b-a)}$$

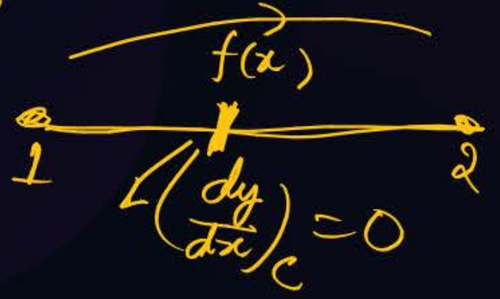
Using Rolles Theorem $f(a)=f(b)$

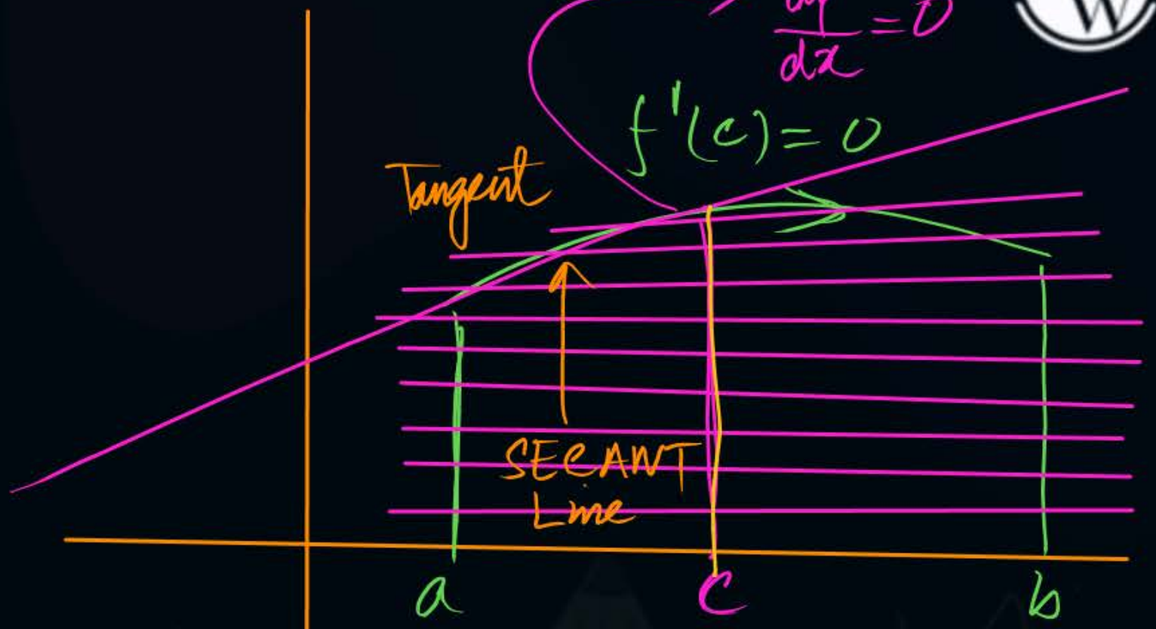
$$\text{slope} = \frac{f(b)-f(b)}{(b-a)} = 0$$

at Pt c

$f'(c)=0$ Rolles Theorem

→ atleast one point $c \in (a,b)$





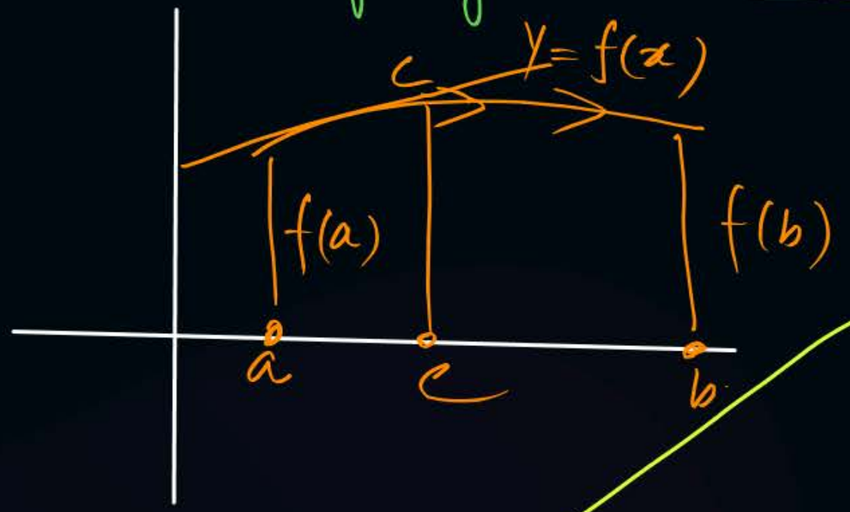
$c \in (a,b)$

$f'(c)=0$ (Rolles Theorem)

$y=f(x)$

5 Rolles Theorem $f(x)_c=0$ 6

Lagrange mean Value Theorem: (LMVT)



$$f'(c) = \frac{f(b)-f(a)}{(b-a)}$$

slope at Pt c

$c \in (a,b)$

open (• •) point Interval
conti [———]

(1) $f(x)$ is Differentiable at (a,b)

2) $f(x)$ is continuous $[a,b]$

3) $f(a) \neq f(b)$

$$f'(c) = \frac{f(b)-f(a)}{(b-a)}$$

$c \in (a,b)$

Necessary and Sufficient Condition for LMVT

Average velocity $= f'(t) = \dfrac{f(t_0)-f(t)}{t_0 - t}$

Differentiability → Property – Point

continuity → Property – Interval

(A) # Differentiable ⟶ must continuous ✓

Point

Interval

(B) # continuous ⟶ Differentiability may or May not be



Topic : Single Variable Calculus

#Q. The number of critical points of $f(x) = \frac{x^2+1}{x^2-x-6}$ is :

$f(x) = \frac{(x^2+1)}{(x^2-x-6)}$

$d\left(\frac{u}{v}\right) = \frac{v\frac{d}{dx}u - u\frac{d}{dx}v}{v^2}$

$f'(x) = \frac{(x^2-x-6)\frac{d}{dx}(x^2+1) - (x^2+1)\frac{d}{dx}(x^2-x-6)}{(x^2-x-6)^2}$

Function is Not defined at Point = Critical Pt

$= 0$ Stationary Pt

$f'(x) = 0$

$x^2-x-6 = 0$

$x^2-3x+2x-6 = 0$

$x(x-3)+2(x-3) = 0$

$x = -2$, $x = 3$

Critical Point

Def. Function Not

A 1

B 2

C 3

D 4



Topic : Single Variable Calculus

H.W

#Q. The number of critical points for the function $f(x) = 6x - 4\cos(3x)$ in $x\epsilon[0,\pi]$ is :

$f(x) = 6x - 4\cos 3x$ $[0, \pi]$

A 1

B 2

C 3

D 4



Topic : Single Variable Calculus

#Q. It is given that for the function $f(x) = x^3 - 6x^2 + ax + b$ on [1,3] Rolle's theorem holds with c=2+(1/√3). Find the values of a and b if f(1) = f(3) = 0.

$f(x) = x^3 - 6x^2 + ax + b$ on [1,3]

Find the values of a and b

Rolles Theorem Holds

$c = 2 + \frac{1}{\sqrt{3}}$

$f(1) = f(3) = 0$

A a=11, b=-6

B a=-11, b=6

C a=11, b=6

D a=-11, b=-6

$f(x) = x^3 - 6x^2 + ax + b$

$f(1) = 0$

$f(1) = 1^3 - 6\times1^2 + a\times1 + b$

$= 1 - 6 + a + b$

$0 = -5 + a + b$

$\boxed{a + b = 5}$ — (1)

$f(3) = 3^3 - 6\times3^2 + 3a + b$

$0 = 27 - 6\times9 + 3a + b$

$= 27 - 54 + 3a + b$

$3a + b = 27$ — (2)

slop ① ✓ Diff ✓ ①

slop ② ✓ conti ✓ ②

slop ③ ✓ $f(a) = f(b)$ ③

slop ④ $f'(c) = 0$ ④

$c = 2 + \frac{1}{\sqrt{3}}$

$a + b = 5$

$3a + b = 27$

$-2a = -22$

$a = 11$

$11 + b = 5$

$11 - 5 = b$

$-6 = b$

$a=11 \quad b=-6$

$f(x)=x^3-6x^2+ax+b$

$f(x)=x^3-6x^2+11x-6$

$f'(c)=0$ using Rolles Theorem

$f'(c)=3c^2-12c+11$

$3c^2-12c+11=0$

$c=2\pm\frac{1}{\sqrt{3}}$

$2+\frac{1}{\sqrt{3}} \in [1,3]$

$c=2+\frac{1}{\sqrt{3}}$ $[1,3]$

$c \in [1,3]$

$2+\frac{1}{\sqrt{3}} \in [1,3]$



Topic : Single Variable Calculus

#Q. If $f(0) = 2$ and $f'(x) = \frac{1}{5 - x^2}$, Then the lower and upper bounds of $f(1)$ Estimated by the mean value theorem are_____.

A 1.9, 2.2

B 2.2, 2.25

C 2.25, 2.5

D None of the above



Topic : Single Variable Calculus

LMVT

Lagrange

#Q. A function $f(x) = 1 - x^2 + x^3$ is defined in the closed interval $[-1, 1]$. The value of x, in the open interval $(-1,1)$ for which the mean value theorem is satisfied, is

A $-\frac{1}{2}$

B $-\frac{1}{3}$

C $\frac{1}{3}$

D $\frac{1}{2}$

$f(x) = 1 - x^2 + x^3 \quad [-1, 1]$

$a = -1 \quad b = 1$

$f(-1) = 1 - (-1)^2 + (-1)^3 = -1$

$f(1) = 1 - (1)^2 + 1^3 = 1$

$$f'(c) = \frac{f(b) - f(a)}{(b-a)}$$

$$= 2c + 3c^2 = \frac{1 - (-1)}{1 - (-1)}$$

$$= \frac{2}{2} = 1$$

$$3c^2 - 2c - 1 = 0$$

Using Lagrange mean value Theorem

$$\rightarrow f'(c) = \frac{f(b) - f(a)}{(b-a)}$$

$$f'(c) = \frac{f(1) - f(-1)}{1 - (-1)}$$

$$3c^2 - 2c - 1 = 0$$

$$c = \frac{+2 \pm \sqrt{4 - 4 \times 3(-1)}}{2 \times 3}$$

$$c = \frac{2 \pm \sqrt{16}}{6} \quad = c = \frac{2+4}{6}$$

$$c = \frac{2 \pm 4}{6} = \frac{2+4}{6} = \frac{6}{6} = 1$$

$$= \frac{2-4}{6} = \frac{-2}{6} = -\frac{1}{3}$$

$$\boxed{c = -\frac{1}{3}}$$

$$c \in [a, b]$$

Basic Integration:

$$\int f(x)\,dx = F(x) + C$$ (Fundamental Theorem of Calculus)

Important Formulae:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

$$\int \sec x \tan x\,dx = \sec x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x$$

$$\int \frac{-1}{1+x^2}\,dx = \cot^{-1} x$$

$$\int \frac{-1}{\sqrt{1-x^2}}\,dx = \cos^{-1} x$$

$$\int e^x\,dx = e^x + C$$

$$\int a^x\,dx = \frac{a^x}{\log a} + C$$

$$\int \frac{1}{x}\,dx = \ln x$$

STRATEGY

Convert to formula list

Using some manipulations



Topic : Problems based on Indefinite Integral

#Q. **Illustration:**

$$\int 2^x.e^x dx$$

Ans.: $\frac{(2e)^x}{\ln(2e)}+c$

$$\int a^x dx = \frac{a^x}{\log_e a}+c$$

$$I = \int 2^x e^x dx \rightarrow \text{convert to The formula List}$$

$$= \int (2e)^x dx = \int (a)^x \quad \text{List of formula}$$

$$= \frac{(2e)^x}{\ln(2e)}+c$$

Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{1-\tan^2 x}{1+\tan^2 x}dx$$

Ans.: $\frac{1}{2}\sin 2x + c$

$\tan x = \frac{\sin x}{\cos x}$ — ①

$$I = \int \frac{1-\tan^2 x}{1+\tan^2 x}dx$$

convert to List of Formula.

$$I = \int \frac{1-\frac{\sin^2 x}{\cos^2 x}}{1+\frac{\sin^2 x}{\cos^2 x}}dx = \int \frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x}dx$$

$$= \int \frac{\cos 2x}{1}dx$$

$$I = \int \cos 2x\, dx$$

Strategy ②

Substitution

$2x = t$

$2dx = dt$

$dx = \frac{dt}{2}$

formula List

$$I = \int \cos t \cdot \frac{dt}{2}$$

$$= \frac{1}{2}\int \cos t\, dt = \frac{1}{2}\sin t + c = \frac{1}{2}\sin 2x + c$$



Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{(\sqrt{x}-1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}}dx$$

$$I=\int \frac{(\sqrt{x}-1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}}dx$$

Convert The formula List

Strategy $\longrightarrow$ manipulation

Ans.:



$$I=\int \frac{(\sqrt{x}-1)\sqrt{x}\,(x^{3/2}-1)}{\sqrt{x}\,(x+\sqrt{x}+1)} \qquad (A^3-B^3)$$

$$=\int \frac{(\sqrt{x}-1)(\sqrt{x}-1)(x+\sqrt{x}+1)}{(x+\sqrt{x}+1)}$$

$$=\int (\sqrt{x}-1)(\sqrt{x}-1)\,dx$$

Convert to formula List

$$=\int (\sqrt{x}-1)^2\,dx$$



Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{\left(x^2+\sin^2 x\right)\sec^2 x}{1+x^2}dx$$

Ans: $\tan x - \tan^{-1} x + c$

Identity

$\sec^2 x = 1+\tan^2 x$

$$I = \int \frac{(x^2+\sin^2 x)\sec^2 x}{(1+x^2)}$$

$$= \int \frac{x^2\sec^2 x + \tan^2 x}{(1+x^2)}$$

$$= \int \frac{x^2(1+\tan^2 x) + \tan^2 x}{(1+x^2)}$$

$$= \int \frac{x^2 + x^2\tan^2 x + \tan^2 x}{(1+x^2)}dx$$

$$= \int \frac{x^2 + (1+x^2)\tan^2 x}{(1+x^2)}$$

$$= \int \frac{x^2 + (1+x^2)\tan^2 x}{(1+x^2)}\, dx$$

$$= \int \frac{x^2}{(1+x^2)} + \int \tan^2 x\, dx$$

$$I = \int \frac{x^2+1-1}{1+x^2}$$

$$= \int \frac{x^2+1}{x^2+1} - \int \frac{1}{1+x^2}$$

$$= x - \tan^{-1} x$$

$$\int \tan^2 x\, dx = \int (\sec^2 x - 1)\, dx$$

$$= \tan x - x$$

$$= x - \tan^{-1} x + \tan x - x$$

$$= \tan x - \tan^{-1} x + C \quad \text{Ans.}$$



Topic : Integral Calculus - I



#Q. **Illustration:**

$$\int \frac{1+\cos^2 x}{1+\cos 2x} dx$$

Ans: $\frac{1}{2}(\tan x + x) + C$



Topic : Integral Calculus - I

H·W

NEST work

#Q. $\int \frac{1+\tan^2 x}{1+\cot^2 x} dx$

Ans.: $\tan x - x + C$



Topic : Integral Calculus - I

#Q. $\int \frac{x^4+x^2+1}{2(1+x^2)}dx$

H.W

Ans.: $\frac{1}{2}\left[\frac{x^3}{3}+\tan^{-1}x\right]+C$



Topic : Integral Calculus - I

#Q. $\int \frac{\sin 2x + \sin 5x - \sin 3x}{\cos x + 1 - 2\sin^2 2x} dx$

H.W

Ans: $-2\cos x + C$

SMC+SMD



Topic : Integral Calculus - I

#Q. $\int \frac{2+3x^2}{x^2(1+x^2)}dx$

H.W

Ans: $-\frac{2}{x}+\tan^{-1}x+C$



Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{\cos(\ln x)}{x} dx$$

Ans: $\sin(\ln x) + c$

$\ln x = t$

$\frac{1}{x} dx = dt$

$= \int \cos t \, dt = \sin t + c$

$= \sin(\ln x) + c$



Topic : Integral Calculus - I

$x^4 = t$

H.W

#Q. **Illustration:**

$$\int \frac{x^3 dx}{1+x^8}$$

Ans: $\frac{1}{4}\tan^{-1}(x^4)+c$



Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{\ln\left(x+\sqrt{1+x^2}\right)}{\sqrt{1+x^2}}dx$$

$\ln(x+\sqrt{1+x^2}) = t$

H.W

Ans: $\frac{1}{2}\left\{\ln\left(x+\sqrt{1+x^2}\right)\right\}^2 + c$



Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{x^2 \tan^{-1} x^3}{1+x^6} dx$$

$x^3 = t$

$x^6 \rightarrow (x^3)^2$

Ans : $\frac{1}{6}(\tan^{-1} x^3)^2 + c$

H.W



Topic : Integral Calculus - I

H.W

#Q. **Illustration:**

$$\int \frac{\tan\sqrt{x}\sec^2\sqrt{x}}{\sqrt{x}}dx$$

Ans: $\left(\tan\sqrt{x}\right)^2 + c$



Topic : Integral Calculus - I

H.W

#Q. **Illustration:**

$$\int \frac{\text{cosec}(\tan^{-1} x)}{1+x^2} dx$$

Ans.: $\ln\left[\frac{1+x^2}{x} - \frac{1}{x}\right] + c$



Topic : Integral Calculus - I

H.W $xe^x = t$

#Q. **Illustration:**

$$\int \frac{e^x(1+x)}{\cos(xe^x)}dx$$

Ans.: $\ln(\sec(xe^x)+\tan(xe^x))+c$



Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{\sin 2x}{\sqrt{9-\sin^4 x}}dx$$

H.W

Ans.: $\sin^{-1}\left(\frac{\sin^2 x}{3}\right)+c$



Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{e^x dx}{\sqrt{e^{2x}-1}}$$

H.W

Ans.: $\ln\left(e^x+\sqrt{e^{2x}-1}\right)+c$



Topic : Integral Calculus - I

#Q. **Illustration:**

$$\int \frac{e^x dx}{4+e^{2x}}$$

H.W

Ans.: $\frac{1}{2}\tan^{-1}\left(\frac{e^x}{2}\right)+c$



2 mins Summary

- Topic One
- Topic Two
- Topic Three
- Topic Four
- Topic Five

✓ Rolles Theorem $f'(c) = 0$ ✓

✓ Lagrange mean value Theorem $f'(c) = \frac{f(b) - f(a)}{(b-a)}$

✓ Basic Integrals

THANK - YOU

